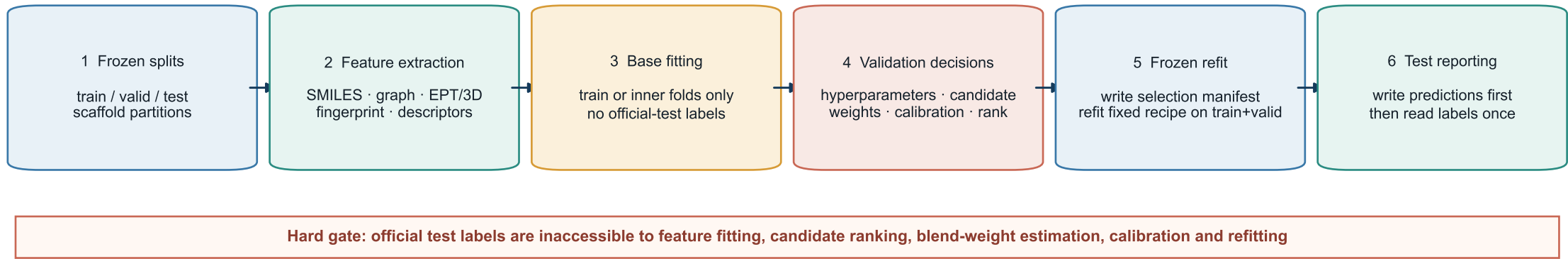


a Leakage-controlled development and evaluation



Family transfer: source-train identities matching target validation/test are removed during selection; source train+valid identities matching target test are removed during final refit

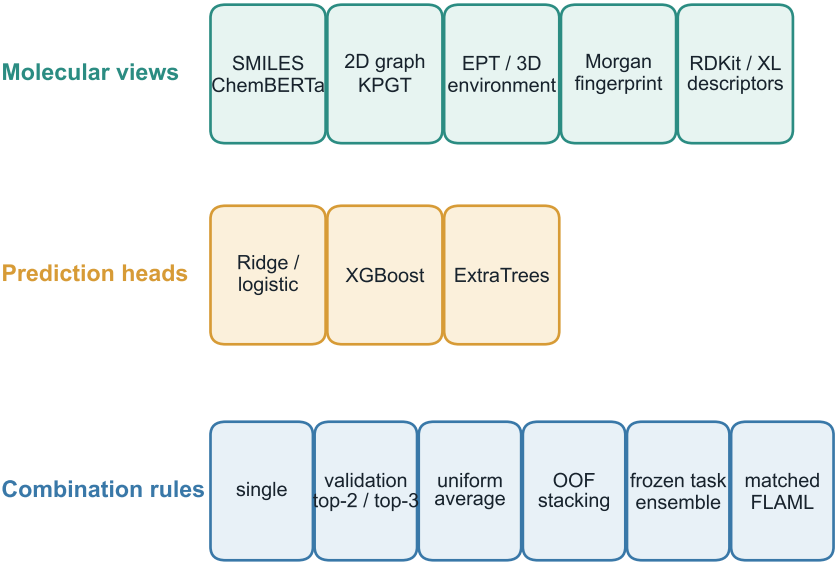
b Twenty-two TDC ADMET tasks

Absorption 6	Distribution 3	Metabolism 6	Excretion 3	Toxicity 4
Caco-2	BBB	CYP2C9-S	CL-hepatocyte	AMES
HIA	PPBR	CYP2D6-S	CL-microsome	DILI
Lipophilicity	VDss	CYP3A4-S	Half-life	hERG
P-gp inhibition		CYP2C9-I		LD50
Solubility		CYP2D6-I		
Bioavailability		CYP3A4-I		

Official metrics: AUROC and AUPRC for classification; MAE and Spearman correlation for continuous endpoints

c Established representations and controlled combination rules

The contribution is the validation-governed task-wise protocol, not a new molecular encoder



Each task receives one machine-readable frozen recipe; all matched controls use the same official splits, representation inputs and five seeds